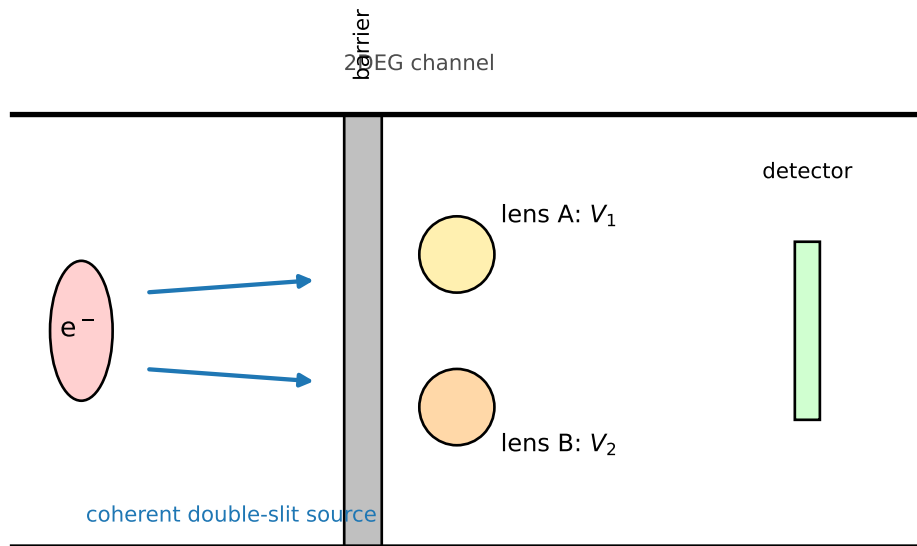


Scalable two-input XOR gate with independent electrostatic lenses

(a) Dual-lens interferometer



$I > I_{th} \Rightarrow$ logic 1

$I < I_{th} \Rightarrow$ logic 0

(b) XOR truth table (shallow-lens regime)

V_1	V_2		out
0	0	balanced	0 / 1 (reset)
0	1	$V_1 < V_2$	0
1	0	$V_1 > V_2$	1
1	1	balanced	0 / 1 (reset)

$\Delta\phi = \phi(V_1) - \phi(V_2)$ linearises the XOR truth table (Eq. 14)